

Data Dashboard Mirage Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	data-dashboard-mirage

Big idea: Dashboards do not simply show data; they prioritize, compress, and narrate it.

Overview

Students compare two dashboards built from the same data and evaluate how interface design shapes conclusions.

Student Prompt

Inspect Data Dashboard Mirage Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Data Dashboard Mirage Stimulus A	Student-facing stimulus 1: Dashboard A highlights falling late arrivals in green; Dashboard B highlights rising absences in amber using the same school-week dataset. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Data Dashboard Mirage Stimulus B	Student-facing stimulus 2: Dashboard A ranks classes by average score; Dashboard B ranks them by improvement. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Data Dashboard Mirage Reveal / Twist	Student-facing stimulus 3: Dashboard A ranks classes by average score; Dashboard B ranks them by improvement. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Dashboards do not simply show data; they prioritize, compress, and narrate it.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Data Dashboard Mirage Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Data Dashboard Mirage Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	When does data visualization become argument?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how dashboards do not simply show data; they prioritize, compress, and narrate it.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use the same fictional dataset in both dashboard versions.
- Change only visual design choices such as color, order, scale, and highlighted metric.

Ready-to-Use Examples

- Dashboard A highlights falling late arrivals in green; Dashboard B highlights rising absences in amber using the same school-week dataset.
- Dashboard A ranks classes by average score; Dashboard B ranks them by improvement.

Suggested Timing

Stage	Teacher move	Watch for
1	Show students two dashboards with the same underlying data but different layouts.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups identify the story each dashboard seems to tell.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students trace which design choices create that impression.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups redesign one dashboard for greater transparency.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief how interfaces shape what users think they know.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- A dashboard answers before we know what question it was built to answer.
- Which design choice did the most interpretive work?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.
- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Support: compare two dashboards with one obvious design difference.
- Challenge: ask students to write a dashboard ethics checklist.

Extension Tasks

- Students redesign a school dashboard for students, parents, and teachers separately.
- Connect dashboard design to public health, climate, or finance communication.

Related Activities

- Graph Caption Framing
- Algorithmic Feed Simulation
- Measurement Changes Phenomenon